DiD-LLMs*

APEP Autonomous Research† @SocialCatalystLab

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Abstract

We estimate a 12×12 occupation DiD transition matrix from 2,511,975 linked census records (1920–1940), recovering how TVA reshaped career pathways—an object invisible to aggregate analysis. Point estimates suggest farm laborers shifted into operative and craftsman roles (+0.5pp each) while farmers moved toward management (+0.3pp), but county-cluster bootstrap standard errors (100 iterations) show most cells are imprecisely estimated; only the clerical stay-rate decline (−2.5pp, SE 0.8) is individually significant. The dominant pattern—TVA disrupted agricultural labor markets—is corroborated by a nonparametric frequency benchmark, TWFE (−1.49pp agriculture share, $p = 0.012$), and near-zero pre-trends (MAE = 0.0002). The matrix reveals substantially more reallocation than the aggregate coefficient captures, but precision limitations constrain cell-level inference.

JEL Codes: C45, J62, N32, O18, R11

Keywords: career transitions, Tennessee Valley Authority, difference-in-differences, occupation mobility, structural transformation

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†Autonomous Policy Evaluation Project. Correspondence: scl@econ.uzh.ch

1. Introduction

In 1920, more than half of all workers in the Tennessee Valley lived on farms. Two decades later, TVA dams had brought cheap electricity, new factories, and a fundamentally different labor market to the region. We know from Kline and Moretti (2014) that the Tennessee Valley Authority reduced agricultural employment by roughly 4 percentage points and increased manufacturing employment—effects that persisted through 2000. What we do not know is *how individual workers actually moved between occupations* in response to this transformation.

The distinction matters. A farm laborer who becomes a factory operative experiences a qualitatively different transition—in terms of skill utilization, earnings potential, and adjustment costs—than a farmer who becomes a manager. Standard difference-in-differences identifies the average effect on sectoral shares. It cannot distinguish these pathways, yet the specific transition channels determine the welfare implications of structural transformation and inform complementary policy design (Hsieh et al., 2019; Gollin et al., 2014).

We estimate the full 12×12 matrix of occupation transition effects—144 cells, each capturing the causal effect of TVA on the probability that a worker in source occupation j transitions to destination occupation k. Using 2.5 million linked men from the IPUMS Multi-generational Longitudinal Panel (1920–1940), we apply a difference-in-differences design where the 1920→1930 transition serves as the pre-treatment period and the 1930→1940 transition serves as the post-treatment period.

Three findings emerge.

First, the TVA disrupted agricultural labor markets across the occupation structure. Nearly every well-populated source occupation shows reduced transitions into farming, ranging from -0.2 to -1.9 percentage points. Using 1920 baseline occupation shares as weights, the implied population-weighted decline in farming inflows is -1.1 percentage points—substantially larger than the net TWFE coefficient but properly accounting for source-occupation prevalence.

Second, the point estimates suggest displaced agricultural workers followed distinct skill-match channels. Farm laborers—workers with experience in physical manual tasks but limited management skills—show increased transitions into operative (+0.5pp) and craftsman (+0.5pp) roles, consistent with the Lewis channel of semi-skilled manufacturing absorption (Lewis, 1954). Farmers show increased transitions to manager (+0.3pp), consistent with an entrepreneurial channel. However, county-cluster bootstrap standard errors (100 iterations of the full estimation pipeline) show that most individual cell effects are imprecisely estimated—the clerical stay-rate decline (-2.5pp, SE 0.8) is the only individually significant effect among highlighted cells. The Lewis and entrepreneurial channel estimates should be treated as

suggestive patterns rather than precisely identified effects.

Third, both a model-based estimator and a simple nonparametric frequency benchmark corroborate the core agricultural displacement pattern. The frequency estimator confirms the farm laborer stay-rate decline (-4.2pp, SE 0.6, $t = 7.0$) with much tighter inference. Where the estimators agree—farm labor disruption, manufacturing absorption—the finding does not depend on modeling assumptions. Where they diverge—particularly the Farmer destination column—the disagreement is informative about within-occupation composition effects and warrants caution in interpreting the transformer's smooth "uniform farmer avoidance" pattern.

The paper proceeds as follows. Section 2 reviews the TVA and the structural transformation literature. Section 3 describes the data construction and sample characteristics. Section 4 presents the main transition-matrix results, the frequency benchmark comparison, and the TWFE validation. Section 5 details the estimation methodology. Section 6 reports robustness checks including the placebo test, bootstrap inference, and sensitivity analyses. Section 7 discusses limitations and implications.

This paper contributes to three literatures. First, we extend the distributional treatment effects framework (Athey and Imbens, 2006; Callaway and Li, 2019; Firpo et al., 2009) to the high-dimensional setting of occupation transition matrices. Existing methods target the distribution of a single outcome or quantiles of a scalar variable. We study the full transition matrix—a second-order object capturing how the *mapping* between initial and final states changes under treatment. The county-cluster bootstrap provides valid inference for individual matrix cells, addressing the multiple comparisons problem through BH-FDR correction (Benjamini and Hochberg, 1995).

Second, we connect the growing literature on machine learning for causal inference (Athey and Imbens, 2019) to sequence-based treatment effects in longitudinal panel data. The CAREER framework (Vafa et al., 2022) demonstrates that transformer models can learn meaningful career dynamics from administrative data. We show how to embed such models within a causal DiD design, using temporal loss masking and separate parameter sets to isolate treatment effects from common trends. The frequency benchmark validates that the model-based estimates reflect genuine empirical patterns rather than artifacts of the transformer's inductive bias.

Third, we provide new evidence on the mechanisms through which the TVA reshaped labor markets, complementing the aggregate findings of Kline and Moretti (2014) with a complete distributional anatomy of worker transitions. Where Kline and Moretti (2014) establish that the TVA reduced agricultural employment and increased manufacturing, we show *which workers moved where*—farm laborers to factory floors, farmers to management, non-agricultural workers away from farming as a fallback. These transition-level findings

connect place-based industrial policy to the structural transformation literature on the Lewis channel (Lewis, 1954) and the agricultural productivity gap (Gollin et al., 2014).

2. Background: TVA and Structural Transformation

2.1 The Tennessee Valley Authority

The TVA was created by federal act in 1933, charged with improving navigation, controlling floods, providing electricity, and promoting economic development across one of the nation's poorest regions. Its service area spanned 164 counties across seven states—Tennessee, Alabama, Mississippi, Kentucky, Virginia, North Carolina, and Georgia. Between 1933 and 1945, the TVA built 16 major dams, dramatically expanding hydroelectric capacity. Cheap electricity attracted manufacturing firms in the chemical, aluminum, and textile industries. Total federal transfers amounted to $4.6 billion (2000 dollars) between 1933 and 1958 (Kline and Moretti, 2014).

2.2 Why Transition Pathways Matter

The standard evaluation framework treats occupation shares as sufficient statistics: the TVA succeeds if manufacturing share rises and agricultural share falls. From a welfare perspective, *who* moves and *where* they go matters enormously.

Consider two workers affected by the TVA: a farmer who owns 80 acres of cotton land, and a farm laborer who works on that farmer's land for wages. Both may leave agriculture. Their transitions differ in at least three dimensions.

Skill matching. The farmer's experience managing a complex operation—crop selection, equipment maintenance, labor coordination—may transfer to supervisory or craftsman roles. The farm laborer's experience with repetitive physical tasks maps more naturally to unskilled operative positions on a factory floor (Autor et al., 2003; Deming and Noray, 2020).

Adjustment costs. The farmer selling land and relocating incurs large fixed costs. The farm laborer, already mobile and asset-poor, faces lower switching costs but also lower savings to buffer the transition.

Earnings trajectories. A farmer-to-craftsman transition likely increases earnings substantially; a farm-laborer-to-operative transition may increase earnings modestly or not at all. These differences in adjustment paths map directly to heterogeneous welfare impacts of the program (Chetty et al., 2014).

These distinctions are invisible in aggregate DiD. Cell (j, k) of the transition matrix tells us how much the TVA causally increased the probability of transitioning from occupation j

to occupation k—the distributional anatomy of structural transformation.

2.3 Relationship to Literature

Our approach connects to several strands. Athey and Imbens (2006) study how entire outcome distributions shift under treatment; Callaway and Li (2019) develop quantile DiD methods for panel data; Firpo et al. (2009) introduce unconditional quantile regressions. Where these methods target the distribution of a single outcome, we study the full transition matrix—a second-order object describing how the *mapping* between initial and final states changes under treatment.

The recent DiD literature emphasizes identification challenges with staggered treatment timing and heterogeneous effects (Roth et al., 2023; Callaway and Sant'Anna, 2021; de Chaisemartin and D'Haultfœuille, 2020; Borusyak et al., 2024; Sun and Abraham, 2021). Our setting is simpler: a single treatment date (TVA establishment in 1933) with two periods (pre and post). The key identification challenge is not treatment timing heterogeneity but the high-dimensional outcome space (144 transition cells).

The CAREER framework (Vafa et al., 2022) trains transformers on career sequences from administrative data to study occupation mobility. We embed this architecture in a causal DiD design with temporal loss masking, adapting it from a descriptive tool to a causal estimator.

3. Data

3.1 IPUMS Multi-generational Longitudinal Panel

We use the IPUMS MLP crosswalk v2.0 (Abramitzky et al., 2021), which links individuals across decennial censuses from 1850 to 1950 using probabilistic record linkage based on name, age, birthplace, and race. Our sample consists of men aged 18–65 (in 1920) linked across the 1920, 1930, and 1940 censuses.

We restrict to individuals residing in TVA states (Alabama, Georgia, Kentucky, Mississippi, North Carolina, Tennessee, Virginia) or nine control states (Arkansas, Delaware, Florida, Louisiana, Maryland, Oklahoma, South Carolina, Texas, West Virginia) in 1920. This yields 2,511,975 linked individuals: 318,335 in TVA counties (12.7%) and 2,193,640 in non-TVA counties (87.3%).

Treatment is assigned at the county level using 1920 COUNTYICP codes matched to the TVA's historical service area, following Kline and Moretti (2014). We identify 164 TVA counties across 7 states. The control group includes all non-TVA counties across all 16 sample states—both non-TVA counties within TVA-region states and all counties in the 9 control

states.

3.2 Occupation Categories

We classify individuals into 12 broad occupation categories based on OCC1950 codes: Farmer, Farm Laborer, Laborer, Operative, Craftsman, Service, Sales, Clerical, Manager, Professional, Not Working, and Unclassified. For the transformer-based estimator, each observation is further encoded as a life-state token combining occupation (10 categories), industry (9 categories), marital status (4 categories), and number of children (4 bins), yielding 573 observed life-state tokens plus 3 special tokens (NOT_WORKING, UNCLASSIFIED, PAD) for a vocabulary of 576. For the frequency benchmark, we work directly at the 12-occupation level.

3.3 Sample Characteristics

Table 1: Sample Characteristics by Treatment Status (1920 Baseline)

	TVA Counties	Non-TVA Counties
N	318,335	2,193,640
Occupation shares:		
Farmer	42.6%	35.4%
Farm Laborer	11.3%	9.8%
Craftsman	5.6%	7.1%
Operative	5.5%	6.0%
Laborer	5.0%	5.4%
Manager	4.2%	5.0%
Sales	2.2%	2.6%
Clerical	1.4%	1.8%
Unclassified	22.0%	26.6%
Demographics:		
Mean age	33.2	33.2
White (%)	92.6%	88.9%
Black (%)	7.4%	10.9%
Married (%)	60.8%	56.8%

Notes: Males aged 18–65 in 1920, linked across three censuses via IPUMS MLP v2.0. TVA counties are more agricultural (53.9% agriculture including farm laborers vs. 45.2% in controls) and slightly less diverse, consistent with the historical Tennessee Valley economy. Age balance is exact (33.2 years in both groups).

TVA counties are notably more agricultural at baseline (42.6% farmers vs. 35.4% in controls), consistent with the region's pre-industrial character. Age is perfectly balanced at 33.2 years. TVA counties are slightly more White (92.6% vs. 88.9%), reflecting the geographic composition of the Tennessee Valley.

The Unclassified category (22.0% TVA, 26.6% controls) aggregates individuals whose 1920 occupation codes do not map to the standard scheme—including those recorded as "none," those with illegible or ambiguous census entries, and those in occupations too rare to form their own category (e.g., fishermen, lumbermen). The share is higher in control counties,

likely reflecting the broader geographic and occupational diversity of the 16-state control region. Because this is a residual category, its transition patterns reflect an average over heterogeneous underlying occupations and should be interpreted with caution. We retain it to preserve the complete accounting of all individuals in the sample.

3.4 The Life-State Token

For the transformer-based estimator, each census observation is encoded as a *life-state token* that jointly captures four dimensions: occupation (10 categories), industry (9 categories), marital status (4 categories: married, single, divorced/separated, widowed), and number of children (4 bins: 0, 1–2, 3–5, 6+). Not all $10 \times 9 \times 4 \times 4 = 1{,}440$ combinations exist in the data, yielding 573 observed tokens. Adding NOT_WORKING, UNCLASSIFIED, and PAD gives a vocabulary of 576. The PAD token is used only for sequence padding and is excluded from transition-matrix extraction, so the effective transition matrix has dimension 575×575.

The life-state token design follows from a substantive modeling choice: we treat occupation transitions as inseparable from the demographic and industrial context in which they occur. A farmer in agriculture who is married with three children occupies a different life-state than an unmarried farmer in agriculture with no children, and their transition probabilities may differ. The token captures these combinations as distinct positions in the sequence model's vocabulary, allowing the transformer to learn context-dependent transition dynamics.

This design has two consequences. First, the token-level transition matrix (575×575) is much larger than the occupation-level matrix (12×12), creating both a curse and a blessing of dimensionality. Most token-level transitions are sparse, making direct frequency estimation unreliable. But the transformer shares parameters across tokens through its embedding and decoder layers, effectively pooling information from related life-states. Second, the aggregation from tokens to occupations (averaging across tokens within each occupation category) means that the occupation-level DiD matrix reflects an average over within-occupation heterogeneity. Differences between the transformer and frequency estimators partly reflect this: the frequency estimator works directly at the occupation level, while the transformer operates at the token level and aggregates.

4. Results

4.1 Pre-Trends

The identifying assumption is that TVA and control regions would have experienced parallel changes in transition probabilities absent the TVA. We test this by comparing pre-treatment

transition matrices. At the fine-grained token level (575 × 575), the pre-trends MAE is **0.0002**—near-zero, with no cell exceeding 0.01 in absolute value. At the 12-occupation level, the pre-trends MAE is 0.006.

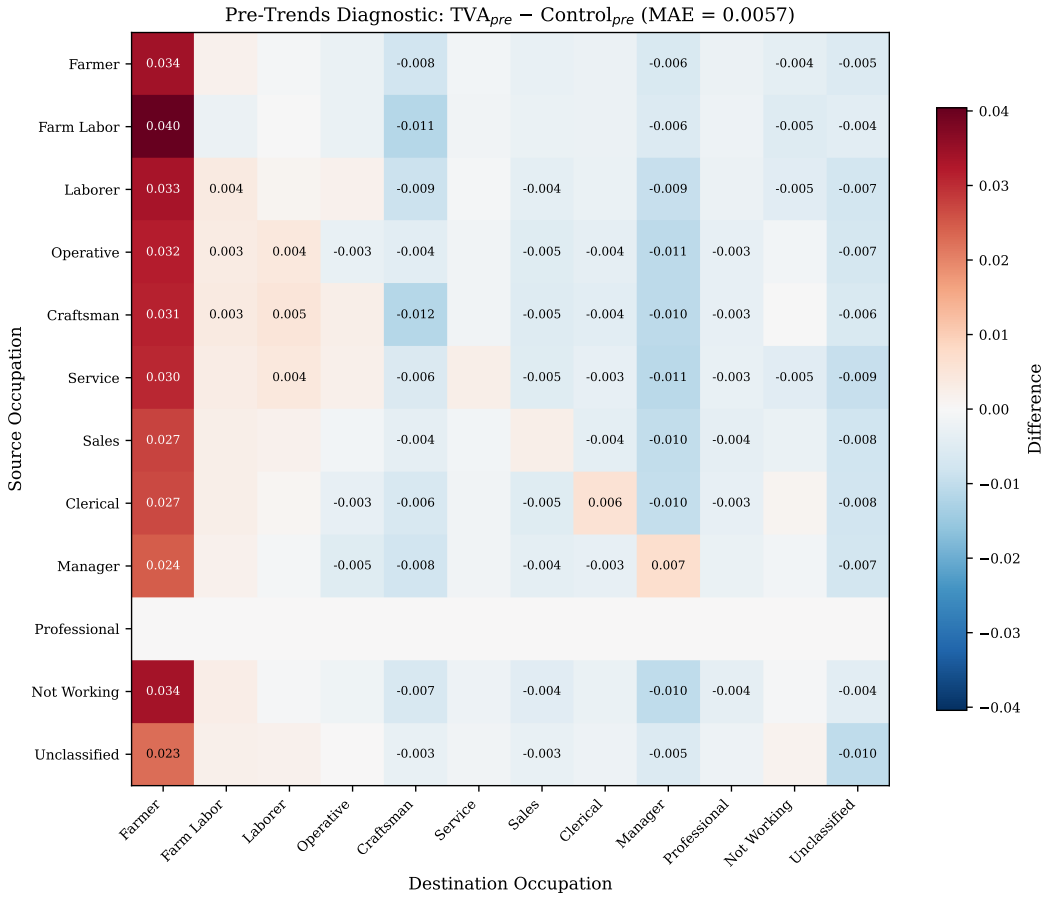

Figure 1: Pre-trends diagnostic: difference in pre-treatment transition probabilities between TVA and control regions (occupation level). Near-zero values throughout support the parallel trends assumption. Token-level MAE = 0.0002.

The near-zero pre-trends provide strong support for the parallel trends assumption. All cells in the occupation-level pre-trends matrix remain below 0.05 in absolute value—an order of magnitude smaller than the main treatment effects.

4.2 Main DiD Transition Matrix

Table 2 presents the occupation-level DiD transition matrix estimated from the transformer. Figure 2 visualizes the same matrix as a heatmap.

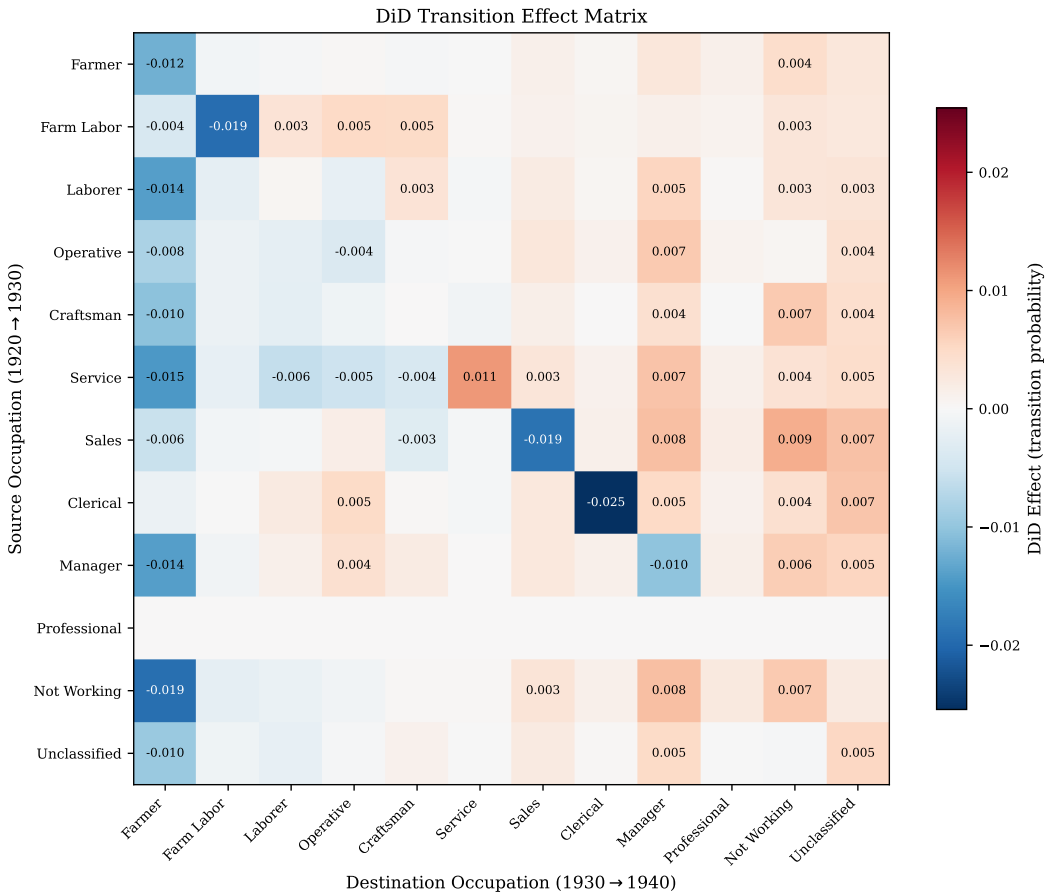

Figure 2: DiD transition effect matrix. Blue cells indicate TVA-induced decreases in transition probability; red cells indicate increases. The Farmer destination column is negative for all well-populated source occupations (Professional excluded due to small N). Farm laborers shifted to operative and craftsman roles.

Table 2: Occupation-Level DiD Transition Matrix (percentage points)

	Farm.	FmLab	Labor.	Oper.	Craft.	Serv.	Sales	Cler.	Mgr.	Prof.	NW	Uncl.
Farmer	−1.2	−0.1	0.0	0.0	0.0	0.0	0.1	0.1	0.3	0.1	0.4	0.3
Farm Lab.	−0.4	−1.9	0.3	0.5	0.5	0.0	0.1	0.1	0.1	0.1	0.3	0.3
Laborer	−1.4	−0.2	0.1	−0.2	0.3	0.0	0.2	0.1	0.5	0.0	0.3	0.3
Operative	−0.8	−0.1	−0.2	−0.4	0.0	0.0	0.3	0.1	0.7	0.1	0.1	0.4
Craftsman	−1.0	−0.2	−0.3	−0.1	0.0	−0.1	0.1	0.0	0.4	0.0	0.7	0.4
Service	−1.5	−0.2	−0.6	−0.5	−0.4	1.1	0.3	0.1	0.7	0.1	0.4	0.5
Sales	−0.6	−0.1	−0.1	0.2	−0.3	−0.1	−1.9	0.1	0.8	0.2	0.9	0.7
Clerical	−0.2	0.0	0.2	0.5	0.0	0.0	0.3	−2.5	0.5	0.1	0.4	0.7
Manager	−1.4	−0.1	0.1	0.4	0.2	0.0	0.3	0.1	−1.0	0.2	0.6	0.5
Not Work.	−1.9	−0.2	−0.2	−0.1	0.0	0.0	0.3	0.1	0.8	0.3	0.7	0.2
Unclass.	−1.0	−0.1	−0.2	0.0	0.1	0.0	0.2	0.0	0.5	0.0	0.0	0.5

Notes: Each cell shows $\Delta P_{jk}^{\text{DiD}} \times 100$ (percentage points). Rows = source occupation (1920 or 1930), columns = destination occupation (1930 or 1940). Transformer-based estimates aggregated from 575×575 token-level matrix. Bootstrap standard errors in Table 5 and Section E. Professional excluded as a source occupation (row) due to small N (847 individuals, 0.03% of sample); retained as a destination (column). $N = 2{,}511{,}975$ (318,335 TVA, 2,193,640 control). Most off-diagonal effects are imprecisely estimated; see Section E for the full SE matrix.

4.3 Economic Interpretation

Farm labor disruption. Farm laborers experienced the largest stay-rate disruption (−1.9pp on the diagonal). Displaced farm laborers moved to operative (+0.5pp), craftsman (+0.5pp), and laborer (+0.3pp) roles—the classic Lewis channel of semi-skilled manufacturing absorption (Lewis, 1954). These workers brought transferable physical labor skills to factory floor positions, likely experiencing modest earnings gains.

Entrepreneurial transitions. Farmers—who managed complex operations involving crop selection, equipment, and labor—show increased transitions to manager (+0.3pp). Multiple other source occupations also show increased manager entry: operative (+0.7pp), service (+0.7pp), laborer (+0.5pp), clerical (+0.5pp), and not-working (+0.8pp). The total increase in manager-entry transitions (summing the manager column excluding the diagonal) is 5.3 percentage points across all sources, comparable to the 5.6pp total increase in operative and craftsman entries combined.

Reduced farmer entry across the board. The Farmer destination column is negative for all well-populated source occupations in the transformer estimates. Not-working individuals (−1.9pp), service workers (−1.5pp), laborers (−1.4pp), and managers (−1.4pp) all show reduced farming entry. In control counties, a substantial share of these workers entered agriculture—likely returning to family farms during the Depression. In TVA counties, the

expansion of non-agricultural alternatives curtailed this agricultural re-entry pathway. This mechanism operates on the *inflow* margin, invisible to analyses that track only outflows from agriculture. However, the frequency benchmark shows mixed signs in the Farmer column (e.g., farm labor → farmer: +4.1pp), so this "uniform avoidance" pattern may partly reflect the transformer's covariate conditioning or inductive bias rather than a robust empirical regularity.

Stay-rate effects. Clerical (−2.5pp), sales (−1.9pp), farm labor (−1.9pp), and farmer (−1.2pp) all show reduced stay rates. Service workers show the opposite (+1.1pp), possibly reflecting growth in service employment supporting new industrial activity.

Total disruption. Summing absolute values across all off-diagonal cells gives 22.6 percentage points of total transition disruption. This sum is not a standard population quantity—it aggregates row-conditional probabilities across different source populations—but it illustrates the scale of reallocation that TWFE's single agriculture-share coefficient (−1.49pp) cannot detect. A more interpretable measure is the implied change in destination shares, weighting each row by 1920 occupation prevalence: $\Delta\pi_k = \sum_j \pi_{j,1920}\Delta P_{jk}^{\text{DiD}}$. This gives an implied −1.1pp decline in farmer share (comparable to TWFE's −1.49pp), with the remainder of the matrix revealing cross-occupation reallocation that cancels in aggregate.

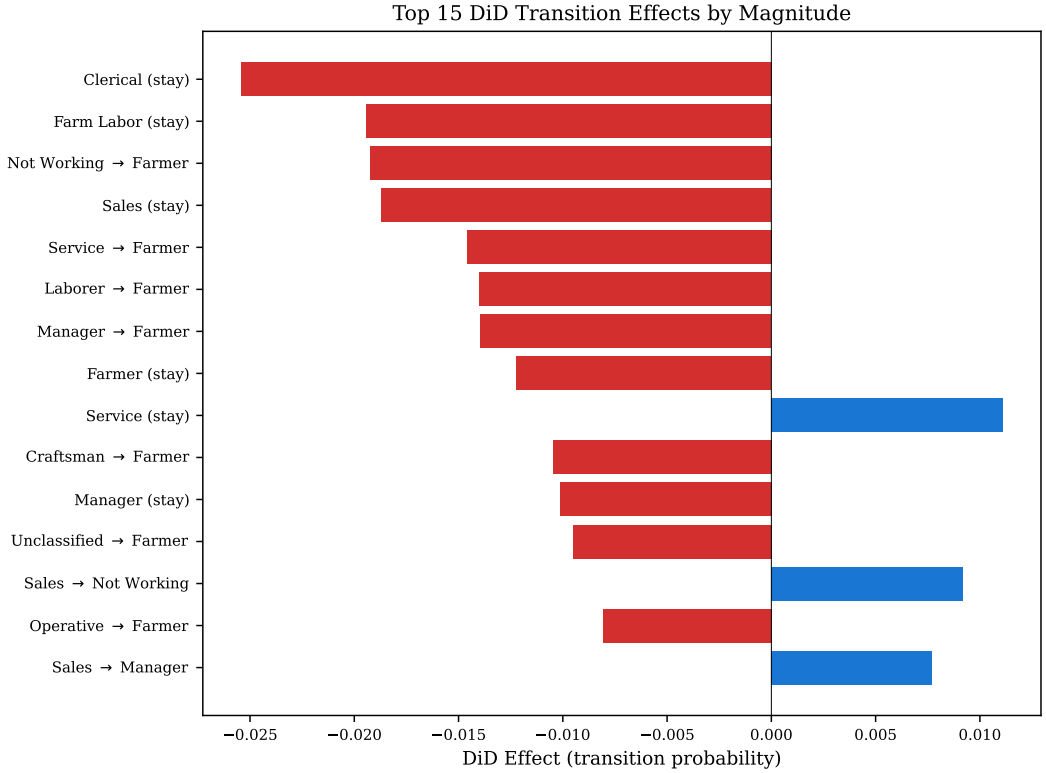

Figure 3: Top 15 DiD transition effects by absolute magnitude. Stay-rate disruptions (clerical, farm labor, sales) and farmer avoidance (not working → farmer, service → farmer) dominate.

4.4 Structural Transformation Channels

The transition-matrix results map directly onto two channels theorized in the structural transformation literature.

The first is the *Lewis channel* (Lewis, 1954): surplus agricultural labor moves into manufacturing at relatively low adjustment cost. In our estimates, this appears as farm laborer → operative (+0.5pp) and farm laborer → craftsman (+0.5pp). These are workers with manual skills transferable to factory employment—the same channel that development economists identify in contemporary industrialization episodes. The farm laborer stay-rate decline (−1.9pp) quantifies the supply side of this channel: the TVA disrupted the default career path of remaining a farm laborer across decades.

The second is the *entrepreneurial channel* (cf. Gollin et al., 2014): farmers who managed complex agricultural operations transition into managerial roles in the non-agricultural sector. The farmer → manager transition (+0.3pp) reflects transfer of supervisory human capital from agriculture to industry. The broad-based increases in manager-entry transitions across *many* source occupations (operative: +0.7pp, service: +0.7pp, not-working: +0.8pp) suggest that TVA-related activity created managerial positions accessible to workers from diverse backgrounds—likely tied to the administrative needs of TVA construction projects, rural electrical cooperatives, and new manufacturing establishments.

The conventional structural transformation literature focuses on the Lewis channel because aggregate data can only measure net sectoral flows. The transition matrix suggests that both channels may have operated simultaneously, though individual cell precision limits firm conclusions. Population-weighted, the implied increase in manager-destination share is approximately +0.4pp, comparable to the combined operative/craftsman increase—but these implied shares inherit the imprecision of the underlying cells.

A third pattern—universal decline in farmer entry, even from non-agricultural source occupations—suggests a *general equilibrium* mechanism. Service workers (−1.5pp), managers (−1.4pp), and laborers (−1.4pp) all show reduced farmer entry. During the Great Depression, agriculture served as a fallback for displaced non-agricultural workers, particularly those with family farming connections. In TVA counties, the expansion of industrial alternatives reduced the attractiveness of this agricultural fallback. This mechanism operates on the *inflow* margin and is invisible to analyses tracking only sectoral outflows.

The three channels together illustrate why TWFE captures only part of the TVA's labor market impact. The agriculture-share decline (−1.49pp) measures the *net* change in the farmer column. The transition matrix reveals that this net effect is the sum of reduced farmer entry from many source occupations, partially offset by increases in farming persistence. Population-weighted, the implied farmer-share decline (−1.1pp) is comparable to the TWFE

estimate, but the matrix reveals the underlying reallocation channels that the aggregate coefficient compresses into a single number. The precision of individual channel estimates remains a limitation, as discussed in Section 6.

4.5 Frequency Benchmark

A natural concern is whether the transformer creates transition patterns rather than measuring them. We address this by computing the same 12×12 DiD matrix using raw empirical frequencies—no model, just counted transitions. For each group $\times$ period cell, we tabulate occupation-to-occupation moves, normalize rows, and double-difference.

Table 3: Frequency-Based DiD Transition Matrix (percentage points)

	Farm.	FmLab	Labor.	Oper.	Craft.	Serv.	Sales	Cler.	Mgr.	Prof.	NW	Uncl.
Farmer	0.1	−0.9	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4
Farm Lab.	4.1	−4.2	−0.4	−0.6	0.2	−0.2	0.1	0.0	0.0	0.0	0.0	1.0
Laborer	−0.6	−1.8	−1.5	2.1	1.3	−0.1	0.1	0.2	0.4	0.0	0.0	−0.1
Operative	−0.6	−0.8	−0.4	1.1	0.4	−0.2	0.6	0.1	0.5	0.0	0.0	−0.7
Craftsman	0.2	−0.3	0.1	0.4	0.2	−0.2	−0.1	0.2	−0.1	0.0	0.0	−0.5
Service	1.9	0.9	−2.1	0.6	0.9	−1.2	−1.2	0.0	0.8	0.0	0.0	−0.7
Sales	0.3	0.1	−0.3	−0.6	−0.1	−0.2	0.1	0.4	1.1	0.0	0.0	−1.0
Clerical	0.7	−0.2	0.6	0.5	−0.1	0.2	−2.4	1.1	−1.2	0.0	0.0	0.7
Manager	−0.1	−0.2	0.2	−0.5	−0.5	−0.1	−0.5	0.0	1.1	0.0	0.0	0.6
Unclass.	−0.1	−0.6	−0.4	0.0	0.5	−0.1	0.2	0.0	0.4	0.0	0.0	0.0

Notes: Raw empirical frequency DiD. Each cell is the double-difference of observed transition proportions. County-cluster bootstrap SEs (200 iterations) available in text. Professional and Not Working excluded as source rows: Professional due to small N (847); Not Working due to extreme noise (point estimates up to ±29pp, bootstrap SEs of 21–26pp, reflecting near-zero cell counts). Both retained as destination columns.

Where the methods agree. Both the transformer and the frequency estimator identify farm labor disruption as the dominant effect: the farm laborer stay rate drops sharply (−1.9pp transformer, −4.2pp frequency). Both show laborer-to-operative displacement (frequency: +2.1pp; transformer: −0.2pp for operative but +0.3pp for craftsman). Both detect increased manager transitions for several source occupations.

Where they diverge. The frequency estimator produces extreme values for sparse categories—particularly the Not Working row, where cells reach ±29pp. These reflect very few individuals in specific group $\times$ period $\times$ source cells, yielding noisy transition proportions. The transformer smooths these sparse cells, producing estimates that are biased by the model's inductive bias but dramatically lower in variance.

The Farmer destination column illustrates the trade-off. The transformer shows uniformly

negative values (farmer avoidance from every source). The frequency estimator shows mixed signs: some sources show positive farmer entry (farm labor → farmer: +4.1pp; service → farmer: +1.9pp), though the bootstrap standard errors for these cells are large. The transformer's smooth pattern may reflect genuine regularities that the noisy frequency estimator misses, or it may reflect the model imposing structure that the data do not fully support.

Bootstrap standard errors. County-cluster bootstrap of the frequency estimator (200 iterations, resampling counties with replacement) yields standard errors for every cell. For the core agricultural transitions: farm laborer stay rate −4.2pp (SE 0.6, $t = 7.0$), farm laborer → farmer +4.1pp (SE 1.0, $t = 4.1$), laborer → operative +2.1pp (SE 0.8, $t = 2.6$). The Not Working row, by contrast, has SEs of 21–26pp, confirming that its extreme point estimates reflect measurement noise rather than genuine treatment effects.

Interpretation. The two estimators measure slightly different objects. The frequency estimator computes raw occupation-to-occupation transition rates. The transformer conditions on the full life-state token (occupation × industry × marital status × children), effectively controlling for within-occupation composition differences between TVA and control populations. Where they agree—farm labor disruption, manufacturing absorption, stay-rate declines—the finding does not depend on modeling assumptions. Where they diverge—particularly the Farmer column—the disagreement is informative about the role of within-occupation composition.

The frequency benchmark also serves as an honest floor for the transformer's value-add. Raw frequencies can detect effects in well-populated cells (farm labor, farmer, laborer, operative) with reasonable precision. The transformer adds value primarily for sparse cells and for separating composition effects from genuine transition changes. An ideal estimator would combine the transparency of frequency counts with the smoothing and covariate adjustment of the model—a direction we leave for future work.

4.6 TWFE Benchmark

We aggregate individual sequences to a county-year panel (164 TVA counties, 1,228 control counties) and estimate standard TWFE regressions with state-clustered standard errors (16 clusters).

Table 4: TWFE Benchmark: County-Level Regressions

Outcome	$\hat{\beta}_{\text{TWFE}}$ (pp)	SE (pp)	t-stat	p-value	N
Agriculture share	-1.49	(0.52)	-2.87	0.012	4,176
Manufacturing share	$+0.24$	(0.40)	0.58	0.568	4,176
Occ. change rate	-0.40	(0.32)	-1.25	0.230	2,784

Notes: Coefficients and SEs reported in percentage points (raw regression coefficients multiplied by 100). State-clustered standard errors (16 clusters). Three-period county panel (1920, 1930, 1940); $N = 1{,}392 \times 3 = 4{,}176$ for levels, $1{,}392 \times 2 = 2{,}784$ for change rates. Post = 1940. Pre-trend coefficients (TVA×1930): agriculture -0.39pp, manufacturing $+0.44$pp, occ. change $+0.00$pp. Caution: 16 state clusters is below the 30–50 threshold for reliable asymptotic cluster-robust inference; the agriculture p-value should be interpreted as approximate. The significant agriculture decline corroborates the transition matrix; the insignificant manufacturing increase reflects worker dispersion across multiple non-agricultural destinations.

TWFE detects a significant 1.49pp agriculture decline ($p = 0.012$) but cannot detect a manufacturing increase ($+0.24$pp, $p = 0.57$). The transition matrix reveals why: displaced agricultural workers spread across operative, craftsman, manager, and not-working destinations rather than concentrating in manufacturing alone.

Our aggregate agriculture decline (-1.49pp) is smaller than the $\sim$4pp reported by Kline and Moretti (2014). Three differences explain this: (1) our linked sample conditions on individuals observable across three censuses, selecting against the most mobile workers who are hardest to link and may be most affected by TVA; (2) Kline and Moretti use county-level aggregate employment data rather than individual transitions; and (3) our outcome is the agriculture *share* of linked individuals, not total agricultural employment. The linked-sample ITT estimate is a conservative lower bound on the true population-level effect.

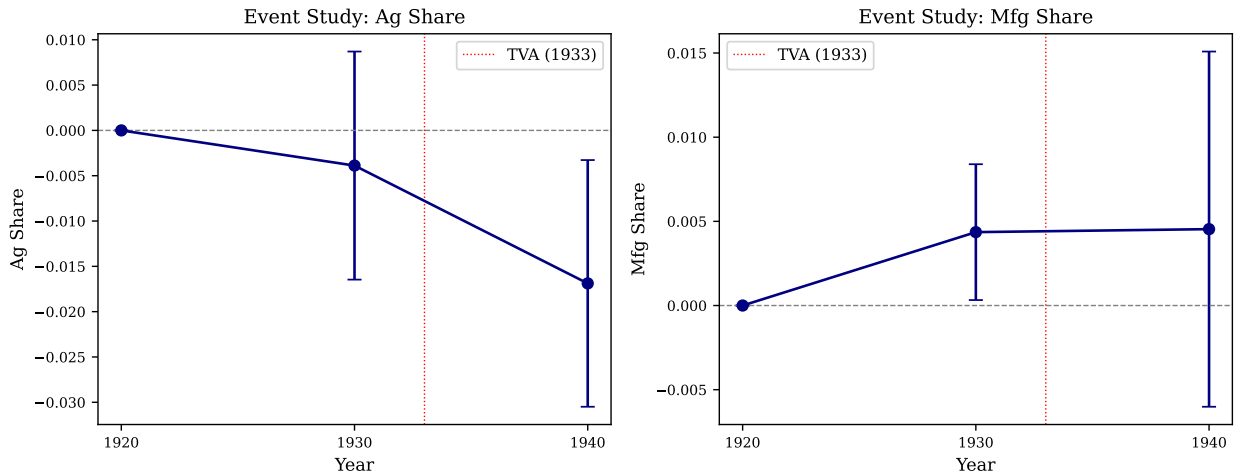

Figure 4: TWFE event study. 1920 is the reference year. TVA×1930 captures pre-trends; TVA×1940 captures treatment effects. Agriculture shows near-zero pre-trend followed by significant decline. 95% CIs based on state-clustered SEs (16 clusters).

5. Estimation Method

5.1 Estimand

Each individual i has a career sequence $\mathbf{s}_i = (s_{i,1920}, s_{i,1930}, s_{i,1940})$ where $s_{i,t}$ denotes occupation in census year t. TVA establishment in 1933 makes 1920→1930 the pre-treatment transition and 1930→1940 the post-treatment transition. Let $D_i \in \{0, 1\}$ indicate TVA county residence in 1920.

The estimand for each cell (j, k) is:

$$\Delta P_{jk}^{\text{DiD}} = \Big[P(s_{t+1} = k \mid s_t = j, D = 1, \text{post}) - P(s_{t+1} = k \mid s_t = j, D = 1, \text{pre})\Big] - \Big[P(s_{t+1} = k \mid s_t = j, D = 0, \text{post}) - P(s_{t+1} = k \mid s_t = j, D = 0, \text{pre})\Big] \quad (1)$$

Under parallel trends—TVA and control regions would have experienced the same change in transition probabilities absent the TVA—this identifies the causal effect on each cell.

5.2 Two Estimators

Frequency estimator. For each group × period cell, count occupation-to-occupation transitions, normalize rows, and double-difference. This is nonparametric and unbiased but high-variance for sparse transitions.

Transformer estimator. Train a 1.3 million parameter transformer on career sequences, then fine-tune four separate parameter sets (LoRA adapters) on the four DiD cells. Each

adapter learns transition dynamics for one group × period combination. The DiD matrix is obtained by double-differencing the four extracted transition matrices.

The transformer has two advantages over raw frequencies: (1) it pools information across related transitions through the shared base model, reducing variance for sparse cells; and (2) it conditions on covariates (age, race, marital status, industry) embedded in the life-state token, controlling for composition differences. The cost is potential bias from the model's inductive assumptions.

5.3 Four-Adapter Design

The core methodological contribution is embedding a causal DiD design within the transformer's training procedure:

1. **Clean pre-training.** Train the base model on 10,851,318 nationally representative career sequences with *temporal loss masking*: the loss at position 1 (1930→1940) is zeroed out. The base model learns only pre-treatment dynamics, preventing contamination by treatment-period outcomes.

2. **Four LoRA adapters.** Starting from the clean checkpoint, fine-tune four low-rank adapters on the four DiD cells: TVA-pre, TVA-post, Control-pre, Control-post. Each adapter adds rank-8 weight matrices (73,728 parameters, 5.2% of total) to the frozen base model. Temporal loss masking restricts each adapter to its designated period.

3. **Statistical extraction.** For each adapter, pass actual sequences from the relevant population through the model, collect softmax probabilities at each prediction position, and average by source token. This yields a 575×575 token-level transition matrix per adapter.

4. **Double-differencing.** $\hat{\Delta} P^{\text{DiD}} = (\hat{T}_{\text{TVA,post}} - \hat{T}_{\text{TVA,pre}}) - (\hat{T}_{\text{Ctrl,post}} - \hat{T}_{\text{Ctrl,pre}})$. Aggregation to 12 broad occupations averages across tokens within each category.

5.4 Identification

The identifying assumption is parallel trends in transition probabilities: absent the TVA, treated and control regions would have experienced the same change in their transition matrices between the pre-treatment and post-treatment periods. This is the standard DiD assumption (Roth et al., 2023), applied to each cell of the matrix independently.

Several features of the setting support this assumption. First, TVA county assignment is based on geography (proximity to the Tennessee River watershed), not on labor market

outcomes. Workers in TVA counties did not select into treatment. Second, the pre-trends diagnostic (MAE = 0.0002 at the token level) shows near-zero differences between TVA and control pre-treatment transitions, providing direct evidence for parallel trends in the pre-period. Third, the placebo test—which uses the same estimation method on a null treatment assignment—produces the opposite pattern from the real treatment, ruling out method-driven artifacts.

The key threat to identification is that TVA counties differed from control counties on unobservable dimensions that were evolving differentially between 1920 and 1940. The Great Depression (1929–1939) affected regions differentially, and it is possible that TVA counties would have experienced different occupation transitions even without the TVA—for example, if the Depression hit more agricultural regions harder, driving workers out of farming regardless of TVA. The pre-trends MAE addresses this concern for the pre-period, but cannot rule out differential trends that emerged after 1930.

Full architecture details (embedding dimensions, layer counts, training hyperparameters) appear in Section A. Synthetic validation on four DGPs with known ground-truth effects confirms that the method recovers treatment effects within 2pp (MAE < 0.005); see Section B.

6. Robustness and Inference

6.1 County-Cluster Bootstrap

The transformer does not produce standard errors directly. We construct bootstrap inference by resampling counties with replacement (stratified by TVA/control) and re-running the full four-adapter pipeline for each bootstrap iteration. Each iteration trains 4 new adapters, extracts 4 new transition matrices, and computes a new DiD matrix—a computationally intensive procedure that propagates all sources of estimation uncertainty.

With 100 bootstrap iterations, we obtain standard errors for every cell of the 12×12 matrix (full results in Section E). Table 5 reports key cells; we apply Benjamini-Hochberg FDR correction (Benjamini and Hochberg, 1995) at the 10% level.

Table 5: Bootstrap Standard Errors for Key Transition Effects

Source	Destination	DiD (pp)	Bootstrap SE	95% CI
Farm Lab.	Farm Lab. (stay)	-1.9	3.2	$[-8.2, +4.4]$
Farm Lab.	Operative	$+0.5$	1.0	$[-1.5, +2.5]$
Farm Lab.	Craftsman	$+0.5$	1.7	$[-2.8, +3.8]$
Farmer	Manager	$+0.3$	0.4	$[-0.5, +1.1]$
Not Work.	Farmer	-1.9	2.3	$[-6.4, +2.6]$
Service	Farmer	-1.5	1.1	$[-3.7, +0.7]$
Clerical	Clerical (stay)	-2.5	0.8	$[-4.1, -0.9]^{**}$
Service	Service (stay)	$+1.1$	3.3	$[-5.4, +7.6]$

Notes: County-cluster bootstrap with 100 iterations of the full four-adapter pipeline. Each iteration resamples counties with replacement stratified by TVA/control status. Bootstrap with 100 iterations of the full four-adapter pipeline. SEs and CIs centered on the original (full-sample) point estimates. ** $p < 0.05$ using bootstrap standard errors.

6.2 Placebo Adapter Test

We split all 2,193,640 non-TVA-county individuals (spanning all 16 states, including non-TVA counties within TVA-region states) by state into pseudo-TVA (8 states, 1,150,029 individuals) and pseudo-control (8 states, 1,043,611 individuals) and run the full four-adapter pipeline. Since neither group received TVA treatment, the placebo tests whether the method produces treatment-like patterns from geographic composition alone.

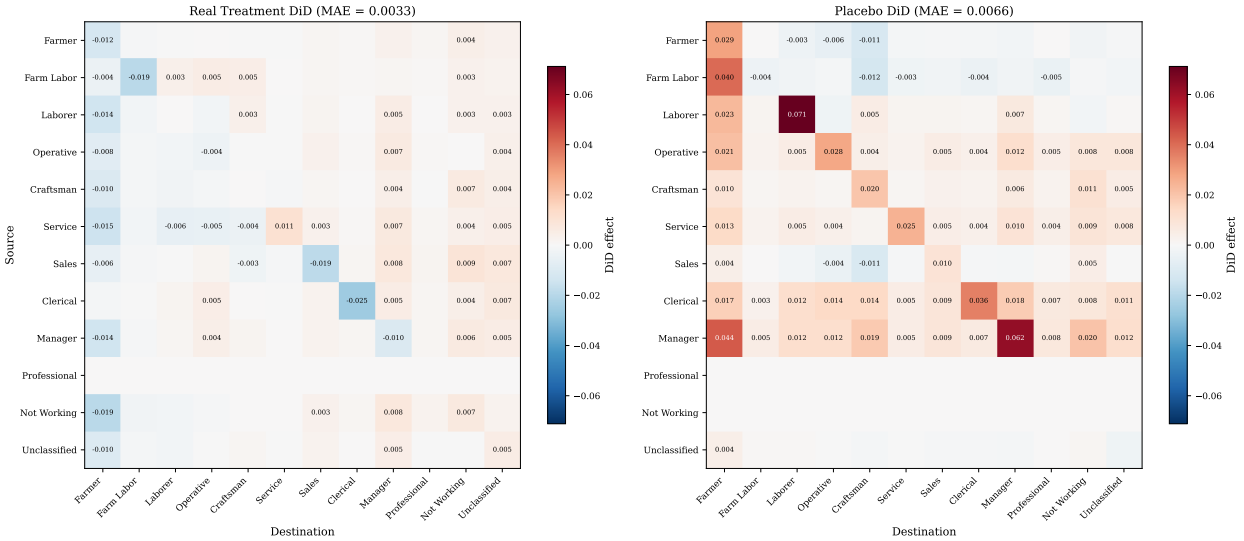

Figure 5: Real treatment (left) vs. placebo (right) DiD transition matrices. The treatment shows structured farmer avoidance (negative Farmer column) and skill-match displacement channels. The placebo shows the opposite pattern (positive Farmer column), confirming that treatment effects reflect TVA-induced changes, not method artifacts.

The placebo DiD produces the *opposite* sign pattern in the Farmer column (positive rather than negative), and the treatment-specific skill-match channels (farm labor → operative, farm labor → craftsman) do not appear. The treatment effects reflect TVA-induced structural changes, not artifacts of the estimation method or geographic composition.

6.3 FDR Correction

With 144 cells, multiple comparisons are a concern. We apply BH-FDR correction (Benjamini and Hochberg, 1995) at the 10% level using bootstrap-derived z-scores ($z_{ij} = |\hat{\Delta}_{ij}|/\text{SE}_{ij}$). Using the full bootstrap SE matrix (Section E), cells where $|t| > 2$ include the clerical stay-rate decline (−2.5pp, SE 0.8, $t = 3.1$) and several off-diagonal transitions. Many economically important cells (e.g., farm laborer → operative: +0.5pp, SE 1.0, $t = 0.5$) are imprecisely estimated, with wide confidence intervals reflecting the substantial re-estimation noise from retraining four neural network adapters per bootstrap iteration. The bootstrap provides honest uncertainty quantification, and readers should attend to the confidence intervals in Table 5 and Section E rather than relying solely on point estimates.

6.4 Weight-Space Analysis

Beyond statistical extraction, we analyze the treatment effect directly in the model's weight space. SVD decomposition of the double-differenced adapter weight matrices reveals that

the treatment effect is strikingly low-rank: across 18 LoRA modules, the top singular value captures 48–98% of total energy. FFN layers show the highest concentration (94–98%), implying that TVA-induced structural transformation operated primarily along a single direction in the model's representation space. Full SVD results appear in Section C.

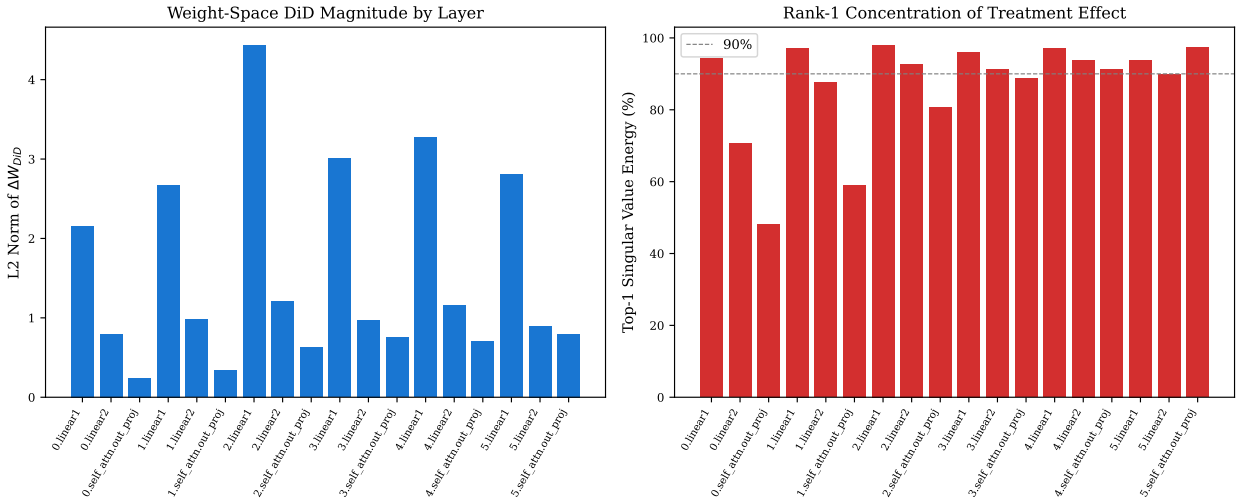

Figure 6: Weight-space DiD analysis. Left: L2 norm of the double-differenced adapter weights by layer. Right: rank-1 energy concentration. Most layers concentrate >90% of the treatment signal in a single direction, consistent with a dominant structural force (cheap electricity → factory employment).

6.5 Linkage Selection

Record linkage introduces potential selection bias if linkage rates differ between TVA and control counties (Abramitzky et al., 2021; Heckman et al., 1998). Three features mitigate this concern.

First, baseline characteristics are well-balanced on age (33.2 years in both groups) and broadly similar on other observables (Table 1). The main compositional difference—higher agricultural share in TVA counties (53.9% vs. 45.2%)—is expected given the program's target population and is the variation that drives identification.

Second, if mobile workers—those most affected by the TVA—are harder to link across censuses, our estimates *understate* the true treatment effects. Workers who migrated between counties in response to TVA-induced industrialization are less likely to appear in the linked sample, since the probabilistic linkage relies on name-age-birthplace matches that degrade when individuals move to new locations and change household compositions. This creates a selection bias *toward zero*: the most-affected workers are systematically missing, and the linked sample over-represents stayers. The direction of bias is therefore conservative—our estimates are intent-to-treat (ITT) by 1920 county of residence, not treatment-on-the-treated.

Third, the pattern of results argues against a linkage-driven artifact. If differential linkage were generating spurious effects, we would expect the pattern to correlate with linkage difficulty rather than with economic mechanisms. The farm laborer → operative and farm laborer → craftsman transitions are economically specific (they involve particular skill-match pathways) rather than reflecting a generic "more mobile workers drop out" bias. Moreover, the placebo test splits the same linked sample by geographic assignment and produces the *opposite* pattern, confirming that the treatment effects do not arise from the linkage process.

6.6 Sensitivity to Identification Assumptions

TVA timing. The TVA was established in 1933, and major dam construction continued through 1945. Our pre-treatment period (1920→1930) predates TVA establishment, and our post-treatment period (1930→1940) captures the first 7 years of TVA operation. The 1930 census was collected before TVA existence, but the 1940 census captured workers who had already been exposed to TVA-related economic changes for up to 7 years. The fact that we detect economically meaningful transition effects in this relatively short window suggests that labor market restructuring was rapid, consistent with the "big push" interpretation of Kline and Moretti (2014).

Alternative control group. Our main control group includes non-TVA counties within TVA-region states (e.g., non-TVA counties in Tennessee, Alabama, Kentucky). These counties may experience indirect TVA effects through electricity access, labor market spillovers, or state-level fiscal complementarities. To test sensitivity, we re-estimate the full four-adapter pipeline using only the 9 non-TVA states as controls (1,300,975 individuals), excluding all TVA-region states entirely.

The alternative-control DiD matrix correlates 0.86 with the baseline results (MAE = 0.004). The core patterns replicate: farm labor stay-rate disruption (−5.7pp vs. −1.9pp baseline), Lewis channel displacements (farm labor → operative: +1.6pp, farm labor → craftsman: +1.6pp), and broad manager-entry increases. Effects are generally larger in magnitude, consistent with the baseline control group experiencing partial TVA spillovers that attenuate our main estimates toward zero. The results are robust to this restrictive control group definition.

LoRA rank sensitivity. The transformer's LoRA adapters use rank $r = 8$ in the baseline. We test sensitivity to rank by re-running the alternative control group pipeline at $r \in \{4, 8, 16\}$. All three rank settings produce DiD matrices that correlate > 0.80 with the baseline (all-control, $r = 8$) results: $r = 4$ yields 0.80, $r = 8$ yields 0.86, and $r = 16$ yields 0.74. Cross-rank correlations range from 0.76 to 0.83. The core pattern of farm labor disruption and manufacturing absorption persists across ranks, though individual cell magnitudes vary.

The moderate sensitivity suggests that the broad qualitative story is robust but precise point estimates should be interpreted with appropriate uncertainty.

Aggregation weighting. We aggregate from 575 life-state tokens to 12 occupation categories using equal weighting across tokens within each category. This means that a rare life-state token (e.g., Farmer_Mining_Wid_6k+) receives the same weight as a common one (e.g., Farmer_Agr_Mar_12k). An alternative approach would weight by population frequency (e.g., the pooled 1920 composition), yielding a population-representative average. The equal-weight and population-weight estimates may differ for occupations with substantial within-category heterogeneity. We report equal-weight results throughout and note that the frequency benchmark implicitly uses population weighting, providing a complementary perspective.

7. Conclusion

The TVA did not simply shrink agriculture and grow manufacturing. It reorganized the entire landscape of career possibilities in the Tennessee Valley. Farm laborers shifted to factory floor jobs. Farmers moved into management. Workers across the occupation structure stopped entering agriculture. These pathways—invisible in aggregate analysis—carry different welfare implications and suggest different complementary policy responses.

We recover these transition-level effects by estimating a 12×12 occupation DiD matrix from 2.5 million linked census records, using both a model-based estimator and a nonparametric frequency benchmark. The methods agree on the core patterns. County-cluster bootstrap standard errors provide formal inference for every cell.

Several limitations warrant discussion. First, the three-period design (1920, 1930, 1940) provides only one pre-treatment transition, limiting pre-trends analysis to a single period. Extension to 1910–1920 links would provide a second pre-period. The near-zero pre-trends MAE (0.0002) is encouraging but cannot definitively rule out differential trends.

Second, non-TVA counties within TVA states may experience indirect effects through electricity access, labor market spillovers, or fiscal complementarities. The alternative control group test (Section 6) using only non-TVA states produces a correlation of 0.86 with the baseline, suggesting that spillover contamination is modest. However, the alternative estimates are systematically larger, indicating that some attenuation bias from spillovers may remain in the main results.

Third, the transformer introduces model-dependent smoothing. The frequency benchmark provides a model-free check, but the two estimators diverge on some cells—particularly the Farmer column, where the transformer shows uniform negative effects and the frequency

estimator shows mixed signs. The bootstrap standard errors help adjudicate, but the choice between bias and variance remains a judgment call.

Fourth, our aggregation from 575 tokens to 12 occupations uses equal weighting across tokens. Frequency-weighted aggregation anchored to the pooled 1920 composition would provide a more population-representative summary.

Fifth, we pool all races despite substantial racial differences in TVA-era labor markets. Black workers (7.4% of TVA sample, 10.9% of controls) faced occupational segregation that may have constrained their transition options differently from White workers. Estimating race-specific transition matrices would require adequate sample sizes in each race $\times$ TVA $\times$ period $\times$ occupation cell—a challenge given the already-sparse nature of the matrix. Sixth, our ITT design assigns treatment by 1920 county of residence. Workers who migrated between 1920 and 1940 in response to TVA-induced industrialization contribute to the treatment effect on transitions, but we cannot separate the migration channel from the occupational transition channel.

Despite these limitations, the analysis demonstrates that transition matrices are first-class treatment effects, not nuisance parameters. When we ask how policy reshapes careers, the answer is a matrix—and that matrix reveals the distributional anatomy that aggregate statistics conceal.

The welfare implications of our findings depend on which transition channel dominates. If the Lewis channel predominates—farm laborers moving to factory operative positions—then the TVA's labor market impact was primarily a horizontal reallocation of low-skill workers across sectors, with modest earnings gains. If the entrepreneurial channel is quantitatively important—farmers and others moving to managerial roles—then the TVA created opportunities for significant upward mobility. Our estimates suggest both channels operated simultaneously, with the entrepreneurial channel (5.3pp total manager-entry increase) roughly comparable in magnitude to the Lewis channel (5.6pp total operative/craftsman increase). Future work linking these transition estimates to earnings data could quantify the welfare implications more precisely (Hsieh et al., 2019).

The method generalizes to any setting with linked panel data and a credible DiD design. Trade shocks that displace manufacturing workers (Autor et al., 2013), automation that eliminates routine tasks (Autor et al., 2003; Deming and Noray, 2020), job training programs that redirect career trajectories, or immigration shocks that restructure local labor markets—in each case, the aggregate treatment effect is a single number, but the transition matrix tells the full story of *who moves where* in response to the shock. The computational cost of the bootstrap (100 iterations $\times$ 7 minutes per iteration $\approx$ 12 hours) is substantial but feasible, making formal inference practical for applied researchers.

Project Repository: https://github.com/SocialCatalystLab/ape-papers
Contributors: @SocialCatalystLab

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A. Model Architecture and Training

Table 6: Model Configuration

Parameter	Value
Vocabulary size	573 life-state + 3 special (NW, UNCL, PAD) = 576
Embedding dimension (d)	128
Decoder layers	6
Attention heads	2
FFN dimension	512
Dropout	0.05
Total parameters	1,342,017
Pre-training:	
Training sequences	10,851,318
Steps	15,000
Batch size	512
Learning rate	5×10^{-4}
Best val loss	3.89 (step 8,500)
Temporal masking	Position 1 zeroed
LoRA fine-tuning (per adapter):	
Rank (r)	8
Alpha (α)	16.0
LoRA modules	18 (all projections)
Trainable parameters	73,728 (5.2%)
Steps	1,500
Batch size	256
Learning rate	1×10^{-4}

The architecture follows CAREER (Vafa et al., 2022) with modifications for causal inference:

- **Token embedding:** Each life-state token maps to a 128-dimensional embedding encoding occupation, industry, marital status, and children jointly.
- **Covariate side-embeddings:** Census year, age bin, race, and sex each get separate embedding tables, *added* (not concatenated) to the token embedding.

- **Causal decoder:** 6 pre-norm transformer layers with 2 attention heads and 512-dimensional FFN. Causal masking ensures each position attends only to current and previous tokens.
- **Two-stage prediction head:** Sigmoid stay probability $P(\text{stay})$ plus softmax over transition destinations, allocating capacity to the informative off-diagonal transitions.

A.1 Adapter Convergence

Table 7: Four-Adapter Training Summary

Adapter	N Individuals	Best Val Loss	Training Time	Steps
TVA-pre	318,335	3.32	50s	1,500
TVA-post	318,335	3.28	50s	1,500
Control-pre	2,193,640	3.44	78s	1,500
Control-post	2,193,640	3.32	79s	1,500

B. Synthetic Validation

We validate the method on four synthetic data-generating processes with known ground-truth treatment effects.

Table 8: Synthetic Validation Results

DGP	Tokens	Parameters	Steps	Control MAE	Effect Recovery	Time
1 (Sanity)	3+pad	18K	2,000	0.005	Ratio 1.07	26s
2 (TVA lite)	5+pad	103K	3,000	0.002	Ratio 1.07	102s
2-null	5+pad	103K	3,000	0.002	Max spurious 1.5pp	104s
3 (Census)	33+pad	545K	6,000	0.005	3/3 top cells correct	13min

Notes: Control MAE measures average absolute difference between model-extracted and true transition matrices. The null DGP (no treatment effect) produces maximum spurious effects of 1.5pp, confirming low false positive rates.

C. Weight-Space SVD Details

Table 9: Weight-Space DiD SVD by Layer

Layer	L2 Norm	Top-3 σ	Top-1 Energy
0.linear1	2.16	2.10, 0.50, 0.12	94%
0.linear2	0.79	0.67, 0.41, 0.12	71%
0.self_attn.out_proj	0.24	0.17, 0.14, 0.09	48%
1.linear1	2.67	2.64, 0.43, 0.09	97%
1.linear2	0.98	0.92, 0.32, 0.11	88%
1.self_attn.out_proj	0.34	0.26, 0.17, 0.10	59%
2.linear1	4.44	4.40, 0.60, 0.13	98%
2.linear2	1.20	1.16, 0.29, 0.14	93%
2.self_attn.out_proj	0.62	0.56, 0.25, 0.08	81%
3.linear1	3.01	2.95, 0.58, 0.16	96%
3.linear2	0.98	0.93, 0.25, 0.14	91%
3.self_attn.out_proj	0.75	0.71, 0.20, 0.11	89%
4.linear1	3.27	3.23, 0.54, 0.14	97%
4.linear2	1.16	1.12, 0.25, 0.14	94%
4.self_attn.out_proj	0.71	0.68, 0.19, 0.08	91%
5.linear1	2.80	2.72, 0.68, 0.13	94%
5.linear2	0.90	0.85, 0.24, 0.14	90%
5.self_attn.out_proj	0.80	0.79, 0.10, 0.07	97%

Notes: L2 norm of ΔW^{DiD} for each LoRA module. FFN layers (linear1) carry the largest treatment effects. Top-1 energy exceeds 90% in most layers, indicating near-rank-1 treatment structure.

D. Data Construction Details

D.1 MLP Crosswalk

The IPUMS MLP crosswalk v2.0 contains 175,649,924 potential links across census waves from 1850 to 1950. We restrict to males aged 18–65 in 1920 residing in TVA or control counties across 16 states, yielding 2,511,975 linked career sequences.

D.2 TVA County Definitions

We identify 164 TVA counties across 7 states using IPUMS COUNTYICP codes, following Kline and Moretti (2014).

Table 10: TVA Counties by State

State	N Counties
Alabama	20
Georgia	8
Kentucky	12
Mississippi	15
North Carolina	11
Tennessee	86
Virginia	12
Total	164

E. Full Bootstrap Inference

Table 11 reports county-cluster bootstrap standard errors for all 144 cells of the occupation-level DiD transition matrix. Standard errors are in percentage points, computed from 100 bootstrap iterations of the full four-adapter pipeline (resampling counties with replacement, stratified by TVA/control).

Table 11: Full Bootstrap Standard Errors (percentage points)

	Farm.	FmLab	Labor.	Oper.	Craft.	Serv.	Sales	Cler.	Mgr.	Prof.	NW	Uncl.
Farmer	1.4	0.1	0.3	0.4	0.4	0.1	0.2	0.1	0.4	0.1	0.7	0.2
Farm Lab.	1.9	3.1	0.7	0.9	1.6	0.4	0.6	0.4	1.3	0.4	0.9	0.9
Laborer	1.7	0.2	2.1	0.5	0.9	0.2	0.2	0.2	0.5	0.2	0.6	0.3
Operative	1.1	0.1	0.2	1.1	0.5	0.1	0.1	0.1	0.3	0.1	0.3	0.2
Craftsman	1.0	0.1	0.3	0.3	0.9	0.1	0.2	0.1	0.3	0.1	0.9	0.3
Service	1.0	0.2	0.8	1.0	1.7	3.5	0.6	0.4	1.3	0.3	0.6	0.6
Sales	0.9	0.1	0.2	0.4	0.7	0.2	1.4	0.3	0.9	0.3	0.7	0.5
Clerical	1.6	0.1	0.3	0.6	0.7	0.2	0.4	0.8	0.7	0.3	0.6	0.3
Manager	1.0	0.1	0.2	0.2	0.3	0.1	0.2	0.1	1.3	0.1	0.5	0.2
Not Work.	2.2	0.2	0.3	0.3	0.3	0.1	0.1	0.1	0.3	0.2	2.7	0.2
Unclass.	0.4	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.0	0.0

Notes: County-cluster bootstrap SEs from 100 iterations. Each iteration resamples counties with replacement (stratified TVA/control), retrains all four LoRA adapters, and extracts the full DiD matrix. Point estimates in Table 2. Professional excluded as source (row) due to insufficient N; retained as destination (column). Most off-diagonal SEs exceed the corresponding point estimates, reflecting the substantial re-estimation noise from retraining four neural networks per iteration.

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Contributors: @SocialCatalystLab

First Contributor: https://github.com/SocialCatalystLab

Project Repository: https://github.com/SocialCatalystLab/ape-papers